Nihal Sanjay Seth

Mechatronics & Robotics Engineer | Robotics · Simulation · Control Systems

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PROFILE

Master’s student in Mechatronics & Robotics (Hochschule Schmalkalden, Germany) with hands-on experience in robot learning, vision-guided robotics, real-time robot control, and simulation. Built FRANK, a three-phase RL and VLA pipeline for Franka Panda in MuJoCo — PPO reach policy, overhead YOLO detection, and language-directed multi-colour cube picking. Built an Autonomous Visual Servoing & Pick-and-Place System on a UR10e (RealSense D435i, YOLO, RTDE) and a Telemetry & Production Intelligence Platform (FastAPI, PostgreSQL, Streamlit). Developed Project AURA — a 6-DOF UR5 dynamics and control framework (MATLAB/Simulink) with trajectory optimization, and a ROS2/Gazebo simulated mobile robot with vision-based autonomous search and object approach. Erasmus+ international project in Finland; engineering leadership in SAE BAJA. Fluent in English; German B1 certified, pursuing B2.

‣ Real-Hardware Robotics	‣ Robotics Simulation	‣ RL & Simulation	‣ Perception & ML
UR10e · RTDE · Visual Servoing	ROS2 · Gazebo · 6-DOF Dynamics	PPO · Curriculum RL · MuJoCo	YOLO · OpenCV · RealSense D435i

EDUCATION

Hochschule Schmalkalden University of Applied Sciences, Schmalkalden, Germany 10/2024 – Present
Master of Engineering – Mechatronics & Robotics

Dwarkadas Jivanlal Sanghvi College of Engineering, Mumbai University, India 07/2019 – 07/2023
Bachelor of Technology – Mechanical Engineering | CGPA: 8.83/10 (German Grade: 1.5)

WORK EXPERIENCE

PMV Electric Pvt Ltd. — Intern, Mechatronic Design Engineer (R&D) 05/2024 – 08/2024

- Designed an electromechanical motor–gearbox unit in SolidWorks, integrating mechanical and drive system requirements for a compact EV platform.
- Performed actuator sizing and drivetrain simulation using ANSYS Motor-CAD and KissSoft, validating motor-drive performance, gear durability, and system efficiency under dynamic load conditions.
- Proposed and developed a reverse differential mechanism, expanding electromechanical system functionality.

DJS Kronos India — Design Head, Combustion 4WD & EV 2WD 03/2020 – 06/2022

- Directed cross-functional mechatronic system development (electric powertrain, suspension, braking) across a 2-year design cycle, coordinating mechanical and electrical subsystems to meet competitive performance targets.
- Led electromechanical powertrain integration for a 2WD Electric ATV, managing motor-drive system coordination and resolving mechanical–electrical interface dependencies across teams.
- Performed simulation-driven structural and drivetrain optimization using Altair Inspire, ANSYS, and SolidWorks; achieved measurable improvements in gearbox efficiency, structural strength, and weight reduction.
- Managed design-to-manufacture workflow covering material selection, iterative design revision, and production planning.

TECHNICAL PROJECTS

Project FRANK – (MuJoCo, PPO, Gymnasium, YOLOv8, OpenCV, PyTorch) 06/2026 – 07/2026
FRANK - Franka Reinforcement and Autonomy via Neural Kinematics

- **Phase 1** — Full RL pipeline for Franka Panda in MuJoCo: custom Gymnasium environment, PPO, curriculum learning, VecNormalize; reach policy drives all arm motion across a 6-stage pick-and-place pipeline; up to 100% success rate across 9 training iterations.
- **Phase 2** — Overhead camera perception via YOLOv8n fine-tuned on 1000 auto-labelled synthetic frames (mAP50 = 0.995, 0.2mm error); perspective projection pixel-to-world transform; arm retracts, detects cube position, executes pick using existing policy.
- **Phase 3 (VLA)** — Three coloured cubes with language instruction: YOLO detects all cubes (single class), HSV classifier identifies target colour, user types instruction in OpenCV window; arm picks only the specified cube sequentially across all three in one session.
- **GitHub:** Project FRANK

Vision-Guided Pick-and-Place & Industrial Monitoring System (UR10e, RealSense D435i, YOLO, FastAPI) 04/2026 – Present
Hochschule Schmalkalden, Germany

Part 1: Autonomous Visual Servoing & Pick-and-Place System

- Built a vision-guided pick-and-place system on a UR10e; trained a custom YOLO object detection model and integrated it with an Intel RealSense D435i for real-time part localization, center estimation, and pose classification.
- Implemented image-space visual servoing for closed-loop TCP alignment using image error feedback and UR Robots RTDE for real-time hardware-level robot control.
- Developed an OpenCV-based orientation estimation pipeline using contour analysis and bounding-box geometry for automatic wrist alignment; integrated adaptive grasp compensation for asymmetric part geometries.

- Integrated conveyor and IR sensor for automated part feeding and count-triggered cycle initiation; upon detecting 4 parts, robot executes the full pick-and-place tray sequence — with ongoing cycle time analysis to benchmark throughput and assess feasibility.

Part 2: Industrial Telemetry & Production Intelligence Platform ("Factory Copilot")

- Developed a modular industrial telemetry platform using FastAPI, PostgreSQL, and Streamlit, enabling real-time monitoring and visualization of robot cycle data, placement statistics, and failed-pick tracking.
- Designed a robot-agnostic signal mapping architecture for standardized telemetry ingestion; implemented anomaly detection, alert generation, and contextual production insights from live robot cycle analytics.
- Architected a distributed monitoring workflow separating robot execution, telemetry ingestion, analytics processing, and dashboard visualization for scalable factory deployment.

Project AURA – Energy-Optimized Robotic Actuation (MATLAB/Simulink, ROS2, Python) 02/2026 – 03/2026

- Built a 6-DOF UR5 dynamic model with FK, Jacobian-based IK, and full inverse dynamics (Coriolis & gravity); optimized trajectories via energy and manipulability metrics with adaptive sampling.
- Implemented singularity-robust Cartesian control using Adaptive Damped Least Squares for stable robot trajectory tracking.
- Designed a payload-aware inverse dynamics framework; quantified 30% increase in actuation demand under 5 kg load.
- Built a modular Simulink architecture integrating trajectory tracking, computed-torque control, and energy/manipulability analysis, with Git-based version control across development iterations (v1.0–v6.0).
- Implemented trajectory optimization using energy and manipulability metrics with adaptive sampling for selecting optimal and feasible robot motion paths.
- Developed a ROS2 pipeline integrating MATLAB-simulated trajectories with RViz visualization, demonstrating the planned architecture for synchronized planner-controller execution with Jacobian-based IK.
- **GitHub Repository:** Project AURA UR5

ROS2 Mobile Robot with Manipulator & Vision-Based Control (ROS2, Gazebo, RViz, Python) 04/2026

- Designed and simulated a mobile robot with a 2-DOF manipulator using ROS2, URDF/Xacro, Gazebo, and RViz.
- Implemented differential drive navigation and joint position control via ROS2 topics and Gazebo plugins.
- Developed a vision pipeline using OpenCV (HSV-based colour segmentation) for real-time object detection.
- Built a modular Python control system using a finite state machine for autonomous search, alignment, and approach behaviors.

Project: Pen Plotter (Master's) 11/2024 – 07/2025

- Designed and built a 3-axis pen plotter frame in SolidWorks for high precision and structural stability.
- Developed electrical schematic in Fritzing and programmed motion control on ESP32 (Arduino IDE, C) with PID tuning.
- Achieved smooth plotting accuracy and reliable homing through tuned PID control and safety integration.
- **GitHub Repository:** Pen Plotter Project

PUBLICATIONS & PATENTS

- Research Paper: "Design and Development of a Quadcopter" — IJRTI, July 2023. (Pick-and-place UAV with custom servo claw, ANSYS-validated structure.) URL: Published Paper (IJRTI 2023)
- Design Patent (Accepted & Published): Continuously Variable Transmission (CVT) for ATV — Application No. 381472-001, Journal No. 01/2024, Date: 05/01/2024. Reduced cost –32%, weight –15% vs. commercial CVT.

TRAINING & INTERNATIONAL EXPERIENCE

SUAS PREMIER – Automation, PLC & Industry Training Programme (Micro-Credential) 05/2025 – 10/2025

Hochschule Schmalkalden, Germany | University-organized industry training programme

- Gained hands-on experience in electronics, microcontrollers (ESP32, Arduino), and PID-based control for automation; completed Siemens TIA Portal PLC training covering ladder/FBD programming, HMI development, simulation, and hardware deployment.

Erasmus+ International Project Course (BIP 2025, Tail Lift System) 08/2025 – 09/2025

LAB University of Applied Sciences, Finland

- Applied SolidWorks CAM, Mathcad & FluidSIM for hydraulic design and manufacturing analysis; modeled and analyzed hydraulic systems in MATLAB/Simulink integrating sensor feedback and control for dynamic performance studies.

SKILLS

Simulation & Middleware	MATLAB/Simulink, ROS2, Gazebo, MuJoCo, UR Robots RTDE
Programming	Python, MATLAB, C/C++, OpenCV, YOLO, FastAPI
Embedded	Arduino, ESP32
CAD	SolidWorks, Inventor, Fusion 360
Tools	Git, Fritzing, Cura, MS Office, PostgreSQL, Streamlit
Languages	English (Fluent) · German B1 certified, actively pursuing B2

RELEVANT CERTIFICATIONS

- MathWorks: MATLAB Onramp · Simulink Fundamentals · Control Design · Image Processing
- ROS 2 for Beginners Level 1 (ROS Jazzy – 2026) · ROS 2 for Beginners Level 2 – TF | URDF | RViz | Gazebo (Udemy – 2026)